📘 DATA ANALYTICS PROJECT DOCUMENTATION

1. Title Page

Project Title: Electronic Vehicles (EV) Search Trend Analysis
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Course: Data Analytics / Data Science
Institution: Inno FortuneIT pvt ltd
Tools Used: Python & Power BI
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2. Abstract

This project focuses on analyzing Electronic Vehicle (EV) search trend data to identify popular keywords and rapidly growing search terms. The dataset consists of 50 EV-related search queries along with their search interest scores and percentage growth values.

The objective of this project is to understand consumer search behavior and detect emerging trends in the EV industry. Data cleaning and preprocessing were performed using Python to handle percentage formatting and ensure data consistency. Exploratory Data Analysis (EDA) was conducted to rank keywords based on popularity and growth rate.

Interactive visualizations and dashboards were created using Power BI to clearly present insights. The results help automobile companies, marketers, and analysts identify high-demand EV models and rising market trends. This analysis demonstrates how search trend data can support data-driven business decisions in the rapidly growing electric vehicle sector.

3. Table of Contents

4. Introduction

The Electric Vehicle (EV) industry is expanding rapidly due to environmental awareness, government incentives, and technological advancements. As competition increases, understanding consumer interest becomes essential for manufacturers and marketers.

Online search trends provide valuable insights into customer preferences, popular models, and emerging keywords. By analyzing EV-related search data, businesses can make informed strategic decisions, optimize marketing campaigns, and forecast demand trends.

This project applies data analytics techniques to extract meaningful insights from EV search trend data.

5. Problem Statement

EV companies and marketers often lack clear visibility into which models or keywords are gaining popularity. Without analyzing search trend data, businesses may:

- Miss emerging high-growth EV models
- Fail to target the right audience
- Lose competitive advantage
- Make uninformed marketing decisions

This project aims to analyze EV search data to identify high-interest and high-growth keywords for better strategic planning.

6. Objectives

- Analyze EV-related search queries
- Identify keywords with highest search interest
- Detect fastest-growing EV search terms
- Compare popularity versus growth trends
- Build an interactive dashboard for visualization
- Provide actionable market insights

7. Dataset Description

- **Source:** Provided CSV dataset
- **File Format:** CSV
- **Total Rows:** 50
- **Total Columns:** 3

Variables Included:

1. **Query** – EV-related search keyword
2. **Search Interest** – Popularity score (0–100 scale)
3. **Increase Percent** – Percentage growth of search interest

The dataset represents trending EV keywords and their growth rates.

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8. Tools & Technologies Used

Category	Tools
Programming	Python
Libraries	Pandas, NumPy
Visualization	Power BI
Data Format	CSV

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9. Methodology (Workflow)

The project followed a structured workflow:

1. Data Collection – Imported CSV dataset
2. Data Cleaning – Removed percentage symbols and converted data types
3. Data Transformation – Standardized column formats
4. Exploratory Data Analysis – Ranked keywords by popularity and growth
5. Visualization – Created charts and KPI cards in Power BI
6. Insight Generation – Identified key trends and emerging keywords
7. Dashboard Development – Built interactive EV Trend Dashboard

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10. Data Cleaning

The following preprocessing steps were performed:

- Removed percentage symbol (%) from “Increase Percent” column
- Converted percentage column to numeric format
- Checked and removed duplicate values
- Standardized column names
- Verified consistency of search interest values

The dataset was clean and required minimal preprocessing.

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11. Exploratory Data Analysis (EDA)

EDA was performed to understand:

- Top keywords by search interest
- Top keywords by growth percentage
- Comparison between popularity and growth
- Identification of emerging EV models

Visualizations used:

- Bar charts
- KPI cards
- Scatter plot (Popularity vs Growth comparison)

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12. Insights & Visualizations

Key findings from the analysis:

- The keyword **“EV”** has the highest search interest score (100).
- Keywords such as **“EV Punch”** and **“Punch EV”** show extremely high growth percentages (250%–300%).
- Some keywords have high growth but moderate popularity, indicating emerging trends.
- Brand and model-specific searches are gaining traction in the EV market.

The Power BI dashboard clearly visualizes these insights using interactive charts.

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13. Results / Output

The project successfully produced:

- Cleaned and structured dataset
- Ranked list of trending EV keywords
- Growth analysis report
- Interactive Power BI dashboard

The dashboard includes:

- KPI Cards (Max Interest, Max Growth, Total Keywords)

- Top Keywords by Popularity Chart
- Fastest Growing Keywords Chart
- Scatter Plot for Trend Analysis

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14. Challenges

- Percentage values stored as text format
- Small dataset size
- Similar keyword variations (e.g., “EV Punch” vs “Punch EV”)
- Limited historical comparison data

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15. Recommendations

- Monitor search trends weekly
- Merge similar keyword variations for deeper insights
- Combine search data with sales data
- Focus marketing efforts on high-growth EV models
- Use forecasting models to predict EV demand

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16. Conclusion

This project successfully analyzed Electronic Vehicle search trend data to identify popular and fast-growing keywords. The analysis highlights rising consumer interest in specific EV models and brand-related searches.

By transforming raw search data into visual insights, this project demonstrates how data analytics can support strategic marketing and business decisions in the rapidly evolving EV industry.

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17. Future Scope

- Integrate Google Trends API for real-time data
- Add multi-year historical comparison
- Implement predictive forecasting models
- Automate Power BI dashboard updates
- Expand analysis to regional segmentation